Lesson 2: Measures and the Carathéodory Extension

AI/ML Mathematics

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1 Motivation and Overview

In the previous lesson we constructed σ-algebras to specify which subsets of a sample space are "measurable." The next step is to assign sizes—lengths, areas, probabilities—to these sets. A *measure* is the mathematical object that does this, and the Carathéodory extension theorem is the fundamental tool for constructing measures rigorously.

The construction matters for two reasons: (1) every probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a measure space, so properties of measures (monotonicity, continuity) become properties of probability used constantly in convergence proofs; and (2) the expected discounted return $\mathbb{E}[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}]$ is an integral with respect to $\mathbb{P}$, requiring the Lebesgue integral built on measures.

Why This Matters for RL

Many RL convergence proofs apply the Borel–Cantelli lemma, the dominated convergence theorem, or Fubini's theorem—all of which require a rigorously constructed measure. The Carathéodory extension theorem guarantees that the probability measure underlying an MDP with continuous state/action spaces actually exists. Without it, expressions like $\mathbb{P}(S_{t+1} \in B \mid S_t = s, A_t = a) = \int_B p(s' \mid s, a)\, ds'$ would be formal manipulations without foundation.

2 Measures

Definition 2.1 (Measure)**.** A *measure* on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu\colon \mathcal{F} \to [0, \infty]$ satisfying:

(i) $\mu(\emptyset) = 0$.

(ii) **Countable additivity:** If $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{1}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is called a *measure space*. If $\mu(\Omega) = 1$, then μ is a *probability measure* and we write $\mathbb{P}$ instead of μ.

Example 2.2 (Standard Measures)**.** (a) **Counting measure:** On any set Ω with $\mathcal{F} = \mathcal{P}(\Omega)$, define $\mu(A) = |A|$ (the cardinality of A, with $|A| = \infty$ if A is infinite).

(b) **Dirac measure:** For $x \in \Omega$, define $\delta_x(A) = \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$

(c) **Lebesgue measure:** On $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the unique measure with $\lambda((a, b]) = b - a$ for all $a < b$. (Existence proved in Section 5.)

(d) **Probability on a finite set:** For $\Omega = \{s_1, \ldots, s_n\}$, any function $p\colon \Omega \to [0, 1]$ with $\sum p(s_i) = 1$ defines a probability measure $\mathbb{P}(A) = \sum_{s_i \in A} p(s_i)$.

Theorem 2.3 (Properties of Measures)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For all $A, B, A_1, A_2, \ldots \in \mathcal{F}$:*

(a) ***Monotonicity:*** $A \subseteq B \implies \mu(A) \le \mu(B)$.

(b) ***Subtractivity:*** $A \subseteq B$ *and* $\mu(A) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$.

(c) ***Countable subadditivity:*** $\mu\big(\bigcup_{n=1}^{\infty} A_n\big) \le \sum_{n=1}^{\infty} \mu(A_n)$.

(d) ***Continuity from below:*** $A_n \uparrow A \implies \mu(A) = \lim_{n\to\infty} \mu(A_n)$.

(e) ***Continuity from above:*** $A_n \downarrow A$ *and* $\mu(A_1) < \infty \implies \mu(A) = \lim_{n\to\infty} \mu(A_n)$.

Proof. **(a)** Write $B = A \sqcup (B \setminus A)$. By countable additivity (applied to a sequence with all but the first two terms equal to $\emptyset$), $\mu(B) = \mu(A) + \mu(B \setminus A) \ge \mu(A)$.

(b) From $\mu(B) = \mu(A) + \mu(B \setminus A)$ and $\mu(A) < \infty$, subtract $\mu(A)$ from both sides.

(c) Define $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. The sets B_n are pairwise disjoint, $\bigcup_n B_n = \bigcup_n A_n$, and $B_n \subseteq A_n$. By countable additivity and monotonicity:

$$\mu\left(\bigcup_n A_n\right) = \mu\left(\bigsqcup_n B_n\right) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n). \tag{2}$$

(d) Suppose $A_n \uparrow A$ (i.e., $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$). Set $A_0 = \emptyset$ and $C_n = A_n \setminus A_{n-1}$. The sets C_n are pairwise disjoint with $A = \bigsqcup_{k=1}^{\infty} C_k$ and $A_n = \bigsqcup_{k=1}^{n} C_k$. By countable additivity:

$$\mu(A) = \sum_{k=1}^{\infty} \mu(C_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mu(C_k) = \lim_{n\to\infty} \mu(A_n). \tag{3}$$

(e) Define $B_n = A_1 \setminus A_n$. Then $B_n \uparrow A_1 \setminus A$. By part (d):

$$\mu(A_1 \setminus A) = \lim_{n\to\infty} \mu(B_n) = \lim_{n\to\infty} \big(\mu(A_1) - \mu(A_n)\big), \tag{4}$$

where the second equality uses part (b) (valid since $\mu(A_n) \leq \mu(A_1) < \infty$). Therefore $\mu(A_1) - \mu(A) = \mu(A_1) - \lim_n \mu(A_n)$, so $\mu(A) = \lim_n \mu(A_n)$. □

Finiteness in Continuity from Above

The condition $\mu(A_1) < \infty$ in part (e) cannot be dropped. Consider counting measure on $\mathbb{N}$ with $A_n = \{n, n+1, n+2, \ldots\}$. Then $A_n \downarrow \emptyset$, but $\mu(A_n) = \infty$ for every n, while $\mu(\emptyset) = 0$. For probability measures, this issue never arises since $\mathbb{P}(\Omega) = 1 < \infty$.

2.1 Null Sets and Complete Measures

Definition 2.4 (Null Set, Complete Measure)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. A set $N \in \mathcal{F}$ is a *μ-null set* if $\mu(N) = 0$. The measure μ is *complete* if whenever N is μ-null and $A \subseteq N$, then $A \in \mathcal{F}$ (and hence $\mu(A) = 0$ by monotonicity).

Proposition 2.5 (Completion of a Measure)**.** *Every measure space $(\Omega, \mathcal{F}, \mu)$ has a* completion*: a complete measure space $(\Omega, \overline{\mathcal{F}}, \bar{\mu})$ with $\overline{\mathcal{F}} \supseteq \mathcal{F}$ and $\bar{\mu}|_{\mathcal{F}} = \mu$, defined by*

$$\overline{\mathcal{F}} = \{A \cup N' : A \in \mathcal{F},\ N' \subseteq N \text{ for some } \mu\text{-null } N \in \mathcal{F}\}, \qquad \bar{\mu}(A \cup N') = \mu(A). \tag{5}$$

Proof. Well-definedness of $\bar{\mu}$. If $A_1 \cup N_1' = A_2 \cup N_2'$ with $N_i' \subseteq N_i$ and $\mu(N_i) = 0$, then $A_1 \triangle A_2 \subseteq N_1 \cup N_2$ (a null set), so $\mu(A_1) = \mu(A_2)$.

$\overline{\mathcal{F}}$ is a σ-algebra. For complement: if $E = A \cup N'$ with $N' \subseteq N$, $\mu(N) = 0$, then $E^c = (A^c \cap N^c) \cup (A^c \cap (N \setminus N'))$; the first part is in $\mathcal{F}$ and the second is a subset of N. For countable union: $\bigcup_n (A_n \cup N_n') = (\bigcup_n A_n) \cup (\bigcup_n N_n')$ with $\bigcup N_n' \subseteq \bigcup N_n$ and $\mu(\bigcup N_n) \leq \sum \mu(N_n) = 0$.

Completeness. If $\bar{\mu}(E) = 0$ with $E = A \cup N'$ and $F \subseteq E$, then $F \subseteq A \cup N$ with $\mu(A \cup N) = 0$, so $F = \emptyset \cup F \in \overline{\mathcal{F}}$. □

3 Algebras and Pre-Measures

To construct measures on σ-algebras (which are typically enormous), we first define them on smaller "algebras" and then extend.

Definition 3.1 (Algebra of Sets)**.** A collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an *algebra* (or *field of sets*) if:

(i) $\Omega \in \mathcal{A}$.

(ii) $A \in \mathcal{A} \implies A^c \in \mathcal{A}$.

(iii) $A, B \in \mathcal{A} \implies A \cup B \in \mathcal{A}$. (closure under *finite* union)

Remark 3.2. An algebra differs from a σ-algebra only in axiom (iii): finite versus countable closure. Every σ-algebra is an algebra, but not conversely.

Example 3.3 (The Algebra of Finite Unions of Half-Open Intervals)**.** On $\mathbb{R}$, the collection $\mathcal{A}$ of all finite disjoint unions of intervals of the form $(a, b]$ (including $\emptyset$ and $\mathbb{R} = (-\infty, \infty)$ as limiting cases) is an algebra but not a σ-algebra. For instance, $\bigcup_{n=1}^{\infty}(0, 1 - 1/n] = (0, 1)$ is not a finite union of half-open intervals, so $(0, 1) \notin \mathcal{A}$.

Definition 3.4 (Pre-Measure)**.** A *pre-measure* on an algebra $\mathcal{A}$ is a function $\mu_0 \colon \mathcal{A} \to [0, \infty]$ satisfying:

(i) $\mu_0(\emptyset) = 0$.

(ii) **Countable additivity on $\mathcal{A}$:** If $A_1, A_2, \ldots \in \mathcal{A}$ are pairwise disjoint and $\bigsqcup_{n=1}^{\infty} A_n \in \mathcal{A}$, then $\mu_0\big(\bigsqcup_n A_n\big) = \sum_n \mu_0(A_n)$.

Example 3.5 (Length as a Pre-Measure)**.** On the algebra $\mathcal{A}$ of Example 3.3, define $\mu_0\big(\bigsqcup_{i=1}^{k}(a_i, b_i]\big) = \sum_{i=1}^{k}(b_i - a_i)$. Finite additivity is immediate. Countable additivity (the non-trivial step) uses the Heine–Borel theorem to show $b - a \leq \sum(b_n - a_n)$ whenever $(a, b] = \bigsqcup_n (a_n, b_n]$; the reverse follows from monotonicity. So μ_0 is a pre-measure.

4 Outer Measures and Carathéodory's Theorem

Definition 4.1 (Outer Measure)**.** An *outer measure* on Ω is a function $\mu^* \colon \mathcal{P}(\Omega) \to [0, \infty]$ (defined on *all* subsets) satisfying: (i) $\mu^*(\emptyset) = 0$; (ii) **monotonicity:** $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$; (iii) **countable subadditivity:** $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Proposition 4.2 (Pre-Measure Induces Outer Measure)**.** *Let μ_0 be a pre-measure on an algebra $\mathcal{A}$. For any $E \subseteq \Omega$, define*

$$\mu^*(E) = \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : A_n \in \mathcal{A},\ E \subseteq \bigcup_{n=1}^{\infty} A_n\right\}. \tag{6}$$

Then μ^* is an outer measure, and $\mu^*(A) = \mu_0(A)$ for all $A \in \mathcal{A}$.

Proof. Property (i): Cover $\emptyset$ by $A_n = \emptyset$ for all n. Then $\sum \mu_0(A_n) = 0$, so $\mu^*(\emptyset) = 0$.

Property (ii): If $A \subseteq B$, every cover of B is a cover of A, so the infimum over covers of A is at most the infimum over covers of B.

Property (iii): Let $E = \bigcup_n E_n$. For each $\varepsilon > 0$ and each n, choose a cover $\{A_{n,k}\}_{k \geq 1} \subseteq \mathcal{A}$ with $E_n \subseteq \bigcup_k A_{n,k}$ and $\sum_k \mu_0(A_{n,k}) \leq \mu^*(E_n) + \varepsilon/2^n$. Then $\{A_{n,k}\}_{n,k}$ covers E, so

$$\mu^*(E) \leq \sum_n \sum_k \mu_0(A_{n,k}) \leq \sum_n \big(\mu^*(E_n) + \varepsilon/2^n\big) = \sum_n \mu^*(E_n) + \varepsilon. \tag{7}$$

Since $\varepsilon > 0$ was arbitrary, $\mu^*(E) \leq \sum_n \mu^*(E_n)$.

μ^* extends μ_0: For $A \in \mathcal{A}$, the single-set cover $A_1 = A$, $A_n = \emptyset$ for $n \geq 2$ gives $\mu^*(A) \leq \mu_0(A)$. For the reverse inequality, suppose $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}$. Define $B_1 = A \cap A_1$ and $B_n = A \cap A_n \cap \big(\bigcup_{k=1}^{n-1} A_k\big)^c$ for $n \geq 2$. Since $\mathcal{A}$ is an algebra, each $B_n \in \mathcal{A}$. The sets B_n are pairwise disjoint with $A = \bigsqcup_n B_n$ and $B_n \subseteq A_n$. By countable additivity of μ_0 on $\mathcal{A}$ and monotonicity of μ_0 (which follows from finite additivity):

$$\mu_0(A) = \sum_n \mu_0(B_n) \leq \sum_n \mu_0(A_n). \tag{8}$$

Taking the infimum over all covers: $\mu_0(A) \leq \mu^*(A)$. $\square$

Definition 4.3 (Carathéodory Measurability)**.** Given an outer measure μ^* on Ω, a set $A \subseteq \Omega$ is *μ^*-measurable* (or *Carathéodory measurable*) if for every $E \subseteq \Omega$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c). \tag{9}$$

Remark 4.4. By subadditivity, $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ always holds. The content of (9) is the reverse inequality: A "splits" every test set E additively.

Carathéodory Extension Theorem

Theorem 4.5 (Carathéodory Extension)**.** *Let μ^* be an outer measure on Ω. Then:*
(a) The collection $\mathcal{M} = \{A \subseteq \Omega : A \text{ is } \mu^*\text{-measurable}\}$ is a σ-algebra.
(b) The restriction $\mu = \mu^*|_{\mathcal{M}}$ is a complete measure on $(\Omega, \mathcal{M})$.
Consequently, if μ_0 is a pre-measure on an algebra $\mathcal{A}$ and μ^* is the induced outer measure (6), then $\mathcal{A} \subseteq \mathcal{M}$ and $\mu|_{\mathcal{A}} = \mu_0$. If μ_0 is σ-finite, the extension to $\sigma(\mathcal{A})$ is unique.

Proof. **Part (a): $\mathcal{M}$ is a σ-algebra.**

Step 1: $\Omega \in \mathcal{M}$. For any E: $\mu^*(E \cap \Omega) + \mu^*(E \cap \emptyset) = \mu^*(E) + 0 = \mu^*(E)$.

Step 2: Closure under complement. The condition (9) is symmetric in A and A^c, so $A \in \mathcal{M}$ if and only if $A^c \in \mathcal{M}$.

Step 3: Closure under finite union. Let $A, B \in \mathcal{M}$. For any test set E, applying the measurability of A and then of B (to the test set $E \cap A^c$):

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).\end{aligned} \tag{10}$$

We need to show $\mu^*(E) = \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Note that $(A \cup B)^c = A^c \cap B^c$, so the third term in (10) is already $\mu^*(E \cap (A \cup B)^c)$. For the remaining two terms: $E \cap (A \cup B) = (E \cap A) \cup (E \cap A^c \cap B)$, so by subadditivity:

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B). \tag{11}$$

Combining with (10):

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c). \tag{12}$$

The reverse inequality holds by subadditivity of μ^*. Therefore $A \cup B \in \mathcal{M}$.

Step 4: Closure under countable disjoint union. Let $A_1, A_2, \ldots \in \mathcal{M}$ be pairwise disjoint. Set $S_n = \bigsqcup_{k=1}^n A_k$ and $S = \bigsqcup_{k=1}^\infty A_k$. By Step 3 and closure under complement (hence under intersection), each $S_n \in \mathcal{M}$.

We prove by induction that for any E:

$$\mu^*(E \cap S_n) = \sum_{k=1}^n \mu^*(E \cap A_k). \tag{13}$$

For $n = 1$: $\mu^*(E \cap S_1) = \mu^*(E \cap A_1)$. For the inductive step, apply the measurability of A_n to the test set $E \cap S_n$:

$$\mu^*(E \cap S_n) = \mu^*(E \cap S_n \cap A_n) + \mu^*(E \cap S_n \cap A_n^c) = \mu^*(E \cap A_n) + \mu^*(E \cap S_{n-1}). \tag{14}$$

Since $S_n \in \mathcal{M}$: $\mu^*(E) = \mu^*(E \cap S_n) + \mu^*(E \cap S_n^c)$. Since $S^c \subseteq S_n^c$, monotonicity gives $\mu^*(E \cap S_n^c) \geq \mu^*(E \cap S^c)$. Using (13):

$$\mu^*(E) \geq \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*(E \cap S^c). \tag{15}$$

Letting $n \to \infty$:

$$\mu^*(E) \geq \sum_{k=1}^{\infty} \mu^*(E \cap A_k) + \mu^*(E \cap S^c) \geq \mu^*(E \cap S) + \mu^*(E \cap S^c), \tag{16}$$

where the last inequality uses subadditivity: $\mu^*(E \cap S) \leq \sum_k \mu^*(E \cap A_k)$. The reverse inequality holds by subadditivity of μ^*. So $S \in \mathcal{M}$.

Step 5: General countable unions. For arbitrary (not necessarily disjoint) $A_n \in \mathcal{M}$, define $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k \in \mathcal{M}$ (by closure under complement and finite intersection). Then $\bigcup_n A_n = \bigsqcup_n B_n \in \mathcal{M}$ by Step 4.

Part (b): μ is a complete measure.

Countable additivity. For pairwise disjoint $A_n \in \mathcal{M}$, set $S = \bigsqcup_n A_n$ and $E = S$ in (13). Letting $n \to \infty$: $\mu(S) = \mu^*(S) \geq \sum_k \mu^*(A_k) = \sum_k \mu(A_k)$. Subadditivity gives the reverse, so $\mu(S) = \sum_k \mu(A_k)$.

Completeness. Suppose $\mu^*(N) = 0$ and $A \subseteq N$. For any test set E: $\mu^*(E \cap A) \leq \mu^*(A) \leq \mu^*(N) = 0$, so $\mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) \leq \mu^*(E)$ by monotonicity (since $E \cap A^c \subseteq E$). The reverse $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ holds by subadditivity. So $A \in \mathcal{M}$.

$\mathcal{A} \subseteq \mathcal{M}$ and uniqueness. For $A \in \mathcal{A}$ and any $E \subseteq \Omega$: let $\{C_n\} \subseteq \mathcal{A}$ cover E with $\sum \mu_0(C_n) \leq \mu^*(E) + \varepsilon$. Then $C_n \cap A, C_n \cap A^c \in \mathcal{A}$ and they cover $E \cap A$ and $E \cap A^c$ respectively. By finite additivity of μ_0: $\mu_0(C_n) = \mu_0(C_n \cap A) + \mu_0(C_n \cap A^c)$. So $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \sum \mu_0(C_n \cap A) + \sum \mu_0(C_n \cap A^c) = \sum \mu_0(C_n) \leq \mu^*(E) + \varepsilon$. Since ε was arbitrary, $A \in \mathcal{M}$.

Uniqueness under σ-finiteness follows from the π-λ theorem (Lesson 1, Corollary): $\mathcal{A}$ is a π-system generating $\sigma(\mathcal{A})$, and two measures agreeing on a π-system with a σ-finite exhaustion agree on the generated σ-algebra. □

5 Lebesgue Measure

Theorem 5.1 (Existence and Uniqueness of Lebesgue Measure)**.** *There exists a unique measure λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\lambda((a,b]) = b - a$ for all $a < b$. This measure is σ-finite and translation-invariant: $\lambda(A + x) = \lambda(A)$ for all $A \in \mathcal{B}(\mathbb{R})$ and $x \in \mathbb{R}$.*

Proof. Let $\mathcal{A}$ be the algebra of finite disjoint unions of half-open intervals $(a_i, b_i]$ (Example 3.3), and let μ_0 be the pre-measure of Example 3.5. The key verification is that μ_0 is countably additive on $\mathcal{A}$.

Suppose $(a,b] = \bigsqcup_{n=1}^{\infty} (a_n, b_n]$ with the union in $\mathcal{A}$. Finite additivity gives $b - a \geq \sum_{n=1}^{N} (b_n - a_n)$ for each N (since $\bigsqcup_{n=1}^{N} (a_n, b_n] \subseteq (a,b]$), hence $b - a \geq \sum_{n=1}^{\infty} (b_n - a_n)$.

For the reverse, fix $\varepsilon > 0$. Consider the closed interval $[a + \varepsilon, b] \subseteq (a,b]$ and the open intervals $(a_n, b_n + \varepsilon/2^n) \supseteq (a_n, b_n]$. These open intervals cover the compact set $[a + \varepsilon, b]$, so by the Heine–Borel theorem, finitely many suffice: $[a + \varepsilon, b] \subseteq \bigcup_{n=1}^{N} (a_n, b_n + \varepsilon/2^n)$. Therefore

$$b - a - \varepsilon \leq b - (a + \varepsilon) \leq \sum_{n=1}^{N} (b_n + \varepsilon/2^n - a_n) \leq \sum_{n=1}^{\infty} (b_n - a_n) + \varepsilon. \tag{17}$$

Since ε was arbitrary, $b - a \leq \sum (b_n - a_n)$.

By Carathéodory's theorem (Theorem 4.5), μ_0 extends to a measure λ on $\sigma(\mathcal{A})$. Since $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$ (the half-open intervals generate the Borel σ-algebra), λ is defined on $\mathcal{B}(\mathbb{R})$.

The extension is unique because μ_0 is σ-finite: $\mathbb{R} = \bigcup_{n=1}^{\infty} (-n, n]$ with $\mu_0((-n,n]) = 2n < \infty$.

Translation invariance: $\mu_0((a + x, b + x]) = b - a = \mu_0((a,b])$, so λ and the shifted measure $A \mapsto \lambda(A + x)$ agree on $\mathcal{A}$. By uniqueness, they agree on $\mathcal{B}(\mathbb{R})$. □

6 Non-Measurable Sets

Theorem 6.1 (Vitali Non-Measurable Set)**.** *There exists a set $V \subseteq [0,1)$ that is not Lebesgue measurable.*

Proof. Define an equivalence relation on $[0,1)$ by $x \sim y$ if and only if $x - y \in \mathbb{Q}$. This partitions $[0,1)$ into equivalence classes (each a coset of $\mathbb{Q}$ intersected with $[0,1)$). By the Axiom of Choice, select exactly one representative from each class; call this set V.

For each $q \in \mathbb{Q} \cap [0,1)$, define the translate $V_q = \{(v+q) \bmod 1 : v \in V\}$, where $x \bmod 1$ denotes the fractional part of x.

Claim 1: The sets $\{V_q\}$ are pairwise disjoint. If $x \in V_{q_1} \cap V_{q_2}$, then $x = (v_1 + q_1) \bmod 1 = (v_2 + q_2) \bmod 1$, so $v_1 - v_2 \in \mathbb{Q}$, hence $v_1 \sim v_2$, giving $v_1 = v_2$ and $q_1 = q_2$.

Claim 2: $[0,1) = \bigsqcup_{q \in \mathbb{Q} \cap [0,1)} V_q$. Each $x \in [0,1)$ satisfies $x \sim v$ for some $v \in V$, and $q = (x - v) \bmod 1 \in \mathbb{Q} \cap [0,1)$ gives $x \in V_q$.

Claim 3: All V_q have the same measure. The map $x \mapsto (x+q) \bmod 1$ preserves λ on $[0,1)$.

Now suppose V were measurable. Then each V_q is measurable with $\lambda(V_q) = \lambda(V) =: c$. By countable additivity:

$$1 = \lambda([0,1)) = \sum_{q \in \mathbb{Q} \cap [0,1)} \lambda(V_q) = \sum_{q \in \mathbb{Q} \cap [0,1)} c. \tag{18}$$

But the right side is 0 if $c = 0$ and ∞ if $c > 0$—neither equals 1. This contradiction shows V is not Lebesgue measurable. □

Remark 6.2. Vitali's theorem shows that no translation-invariant measure can be defined on all of $\mathcal{P}(\mathbb{R})$—the fundamental reason for restricting to σ-algebras. In RL with continuous state spaces, all practically arising sets are Borel, but Vitali's result justifies the measure-theoretic formalism.

7 Exercises

Exercise 7.1 (Continuity of Measures)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $(A_n)_{n \geq 1}$ be a sequence in $\mathcal{F}$.

(a) Prove that $\mu\big(\liminf_n A_n\big) \leq \liminf_n \mu(A_n)$, where $\liminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$.

(b) Show by example that the inequality can be strict.

(c) If $\mu(\Omega) < \infty$, prove $\mu\big(\limsup_n A_n\big) \geq \limsup_n \mu(A_n)$, where $\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$.

Hint: Define $B_n = \bigcap_{k=n}^{\infty} A_k$ and apply continuity from below.

Exercise 7.2 (Verifying Outer Measure Properties)**.** Let $\Omega = \{1,2,3\}$ and define $\mu^* \colon \mathcal{P}(\Omega) \to [0,\infty]$ by $\mu^*(\emptyset) = 0$, $\mu^*(\{i\}) = 1$ for each i, $\mu^*(\{1,2\}) = 1$, $\mu^*(\{1,3\}) = 2$, $\mu^*(\{2,3\}) = 2$, and $\mu^*(\{1,2,3\}) = 2$.

(a) Verify that μ^* is an outer measure.

(b) Determine which subsets of Ω are μ^*-measurable (i.e., satisfy the Carathéodory condition for *every* test set E).

(c) What is the resulting σ-algebra $\mathcal{M}$, and what is $\mu = \mu^*|_{\mathcal{M}}$?

Exercise 7.3 (Borel–Cantelli Application)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a measure space with $\mu(\Omega) < \infty$. Using only the properties proved in Theorem 2.3, prove:

(a) If $\sum_{n=1}^{\infty} \mu(A_n) < \infty$, then $\mu(\limsup_n A_n) = 0$.

(b) Interpret part (a) when $\mu = \mathbb{P}$ is a probability measure: "if the total probability is summable, then the event that infinitely many A_n occur has probability zero."

This is the first Borel–Cantelli lemma, which is fundamental in RL convergence proofs.

Hint: Use subadditivity on $\bigcup_{k=n}^{\infty} A_k$ and then continuity from above.

Exercise 7.4 (Completion of a Measure)**.** Let $\Omega = \{1, 2, 3\}$ with $\mathcal{F} = \big\{\emptyset, \{1\}, \{2, 3\}, \Omega\big\}$ and $\mu(\emptyset) = 0$, $\mu(\{1\}) = 1$, $\mu(\{2, 3\}) = 2$, $\mu(\Omega) = 3$. Is μ complete? Compute the completion $(\Omega, \overline{\mathcal{F}}, \bar{\mu})$ explicitly.

References